IDEATE PHASE

Architectural Digital Twin cum Simulator

1. Introduction

The Ideate phase focuses on generating possible solutions to the problem identified during the Define phase. For this project, the team explored ways to help architecture and civil engineering students explore building designs, compare design alternatives, and understand simulation information in one interactive environment. Ideas were generated before selecting a direction for further development.

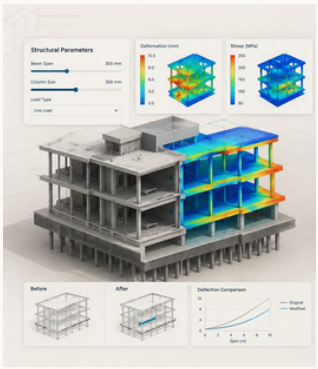
2. Brainstorming

To generate possible solutions by identifying the research gaps found in the literature survey on digital twins, building simulation, material information, and architectural education.

S. No.	Idea	Description
1	**Structural Behaviour Simulator**	Allow users to modify selected structural parameters and explore simulated responses under defined conditions.
2	**Smart Material Comparison**	Compare building materials based on properties such as density, thermal conductivity, durability, and cost estimates.
3	**Sunlight and Heat Exposure Simulator**	Visualise how building orientation, window placement, and shading affect sunlight exposure.
4	**Rainwater Flow Simulator**	Visualise estimated rainwater flow across roofs, gutters, and drainage paths.

5	**Interactive Design Consequence Explorer**	Allow users to modify a building model and compare the original and modified designs.
6	**Building Component Information Explorer**	Select building components to view their dimensions, materials, and relevant properties.
7	**Design Version Comparison Dashboard**	Compare different building design versions using supported parameters and visual comparisons.
8	**Guided Architecture Learning Mode**	Provide small design challenges that help students explore building design concepts interactively.

3. Brain writing



Concept: Let users modify a building's shape or selected structural parameters and visualise how the model responds under defined loads.

Example: Change the span or height of a modelled beam and compare the calculated deformation.

Problem addressed: Beginners struggle to understand how design changes affect structural behaviour.

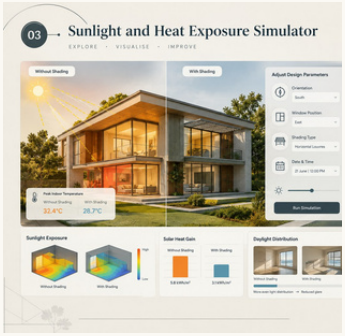
Unique feature: A visual before-and-after comparison with deformation indicators.



Example: Compare timber, steel and concrete using stored properties such as density, cost assumptions and thermal conductivity.

Problem addressed: Choosing materials involves balancing several competing factors.

Unique feature: A side-by-side comparison of materials and their effects on selected performance indicators.



Concept: Estimate how building orientation, window placement and shading affect solar exposure.

Example: Compare a room with a large unshaded window against the same room with an external shade.

Problem addressed: Building designs can create unwanted solar exposure and thermal discomfort.

Unique feature: A visual solar-exposure map for different design alternatives.



Idea 4 — Rainwater Flow Simulator

Concept: Visualise how rainfall could flow across a roof and toward gutters and drainage points.

Example: Change the roof slope and compare the estimated runoff paths.

Problem addressed: Poor roof drainage design can contribute to water accumulation and leakage.

Unique feature: Animated water-flow paths and potential drainage problem indicators based on a defined model.



Example: Increase a cantilever's length and display the corresponding calculated change in deformation under specified assumptions.

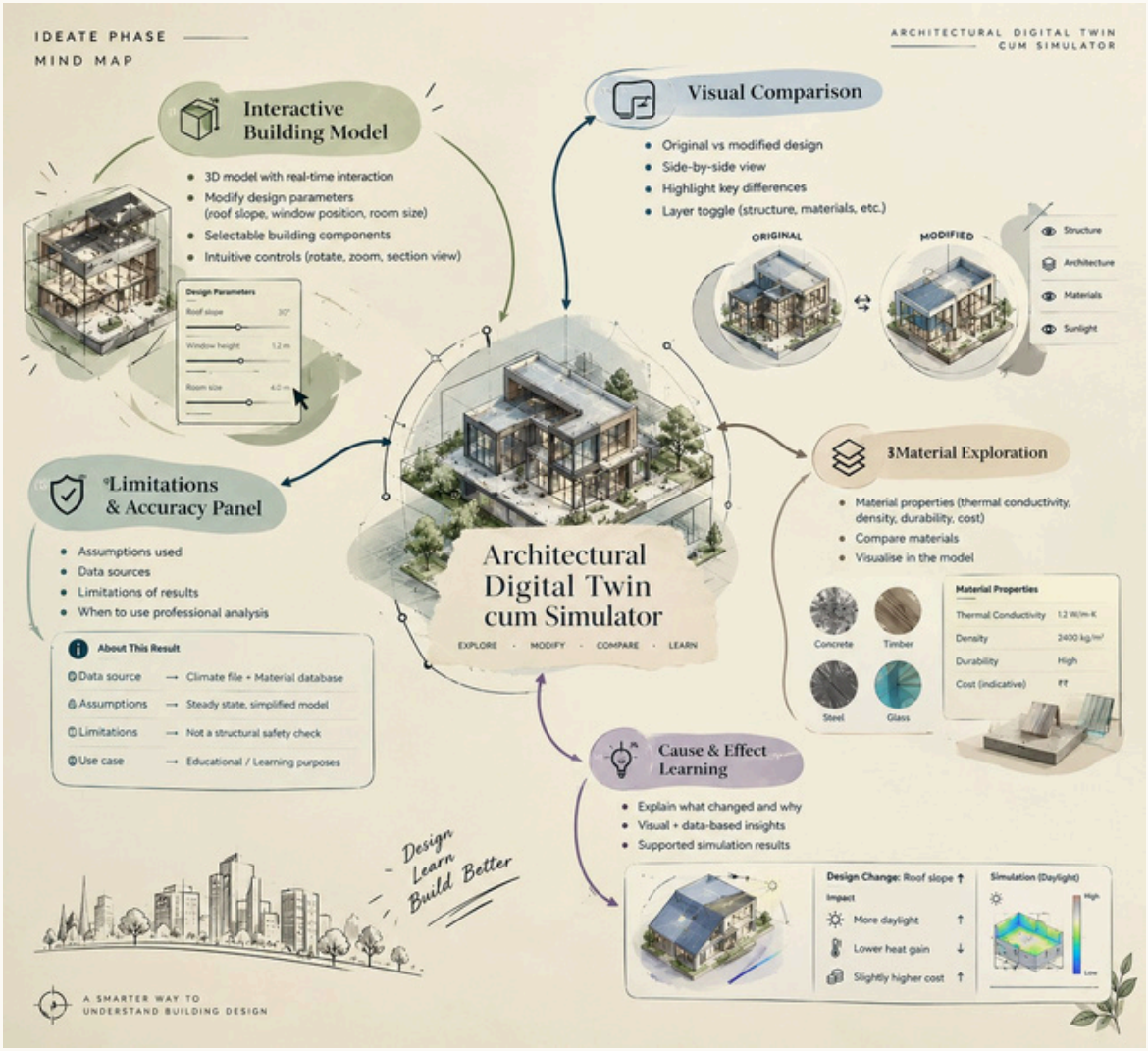
Problem addressed: Users may see a design but struggle to understand the consequences of modifying it.

Unique feature: An interactive simulation mode that connects design changes to calculated outcomes.

4. Challenge Assumptions

Assumption	Challenge	Revised idea
Students can understand building behaviour from static drawings alone	Static drawings may not clearly communicate how a design changes when parameters are modified.	Introduce interactive model modification and visual comparisons.
A 3D model is sufficient for understanding a building.	A model may show appearance without explaining material properties or calculated performance	Connect the 3D model with relevant component information and supported simulation outputs.
Material selection depends mainly on appearance.	Different materials have different physical properties, costs, and practical limitations.	Add a material comparison feature using documented data.
Every design change can be evaluated through visual inspection.	Some consequences require calculations or specialised simulation methods	Separate visual changes from calculated results and clearly state the model's limitations.
A complex professional digital twin is necessary for every learning activity.	Students may benefit from a smaller, focused model designed around specific learning objectives	Build a lightweight, student-centred interactive simulator.
More simulation features always produce a better learning tool.	Too many features may make the prototype difficult to understand and develop	Prioritise a few useful, feasible features and expand only when justified.

5. Mind Map



6. Selected Ideation Direction

Selected Concept: Architectural Digital Twin cum Simulator

The proposed solution combines the most relevant elements into a focused learning environment where users can explore a building model, modify selected parameters, compare design versions, and examine material information and supported simulation outputs.

The main focus is not to invent digital twins. It is to explore whether a simplified, student-centred experience can make design consequences easier to understand.

7. Conclusion

The Ideate phase produced a range of possible features and challenged assumptions about student learning needs. The resulting direction combines an interactive building model with controlled design modifications, visual comparison, and understandable simulation information. This concept can be refined and tested in the next stage of the Design Thinking process.